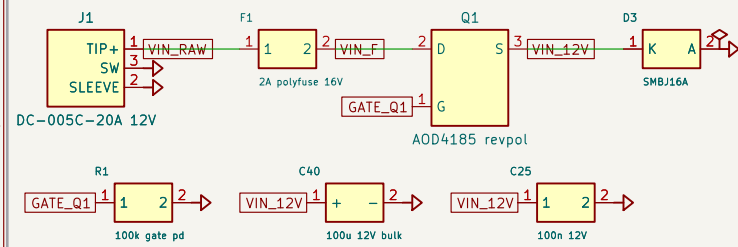
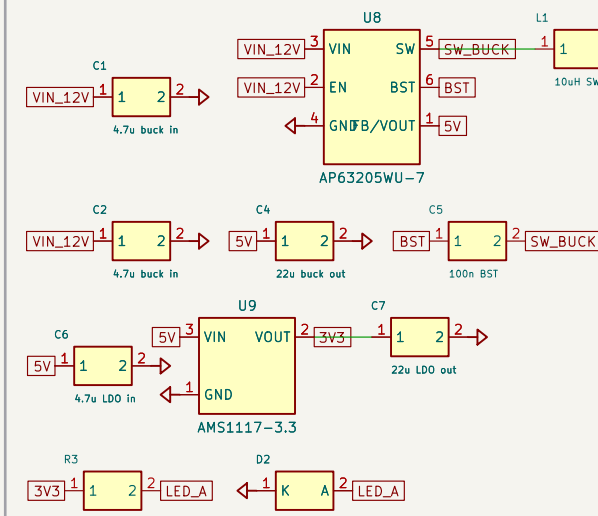


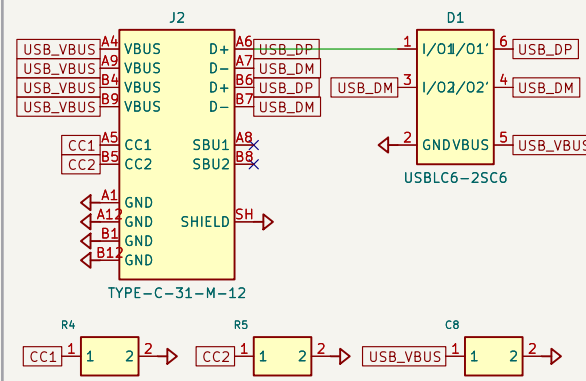
1. 12V INPUT: barrel -> 2A polyfuse -> P-FET revpol -> SMBJ16A TVS [ADR-0001]



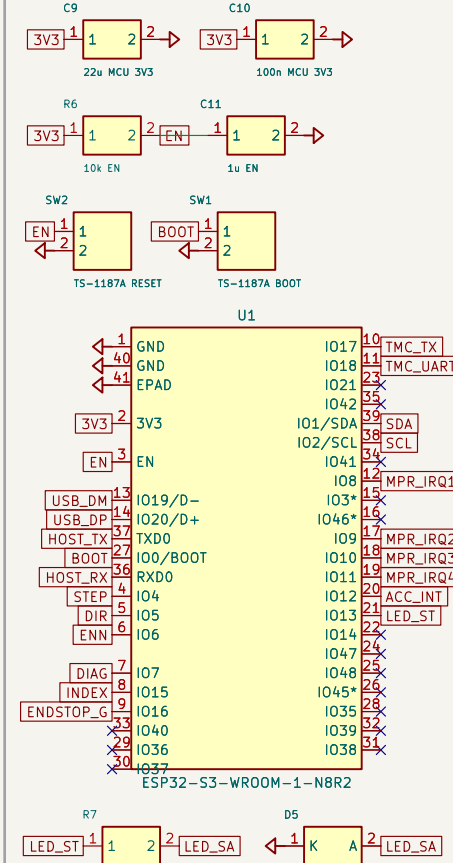
2. POWER: AP63205 12V->5V 2A (host 1.5A max); AMS1117 -> 3V3 [D5]



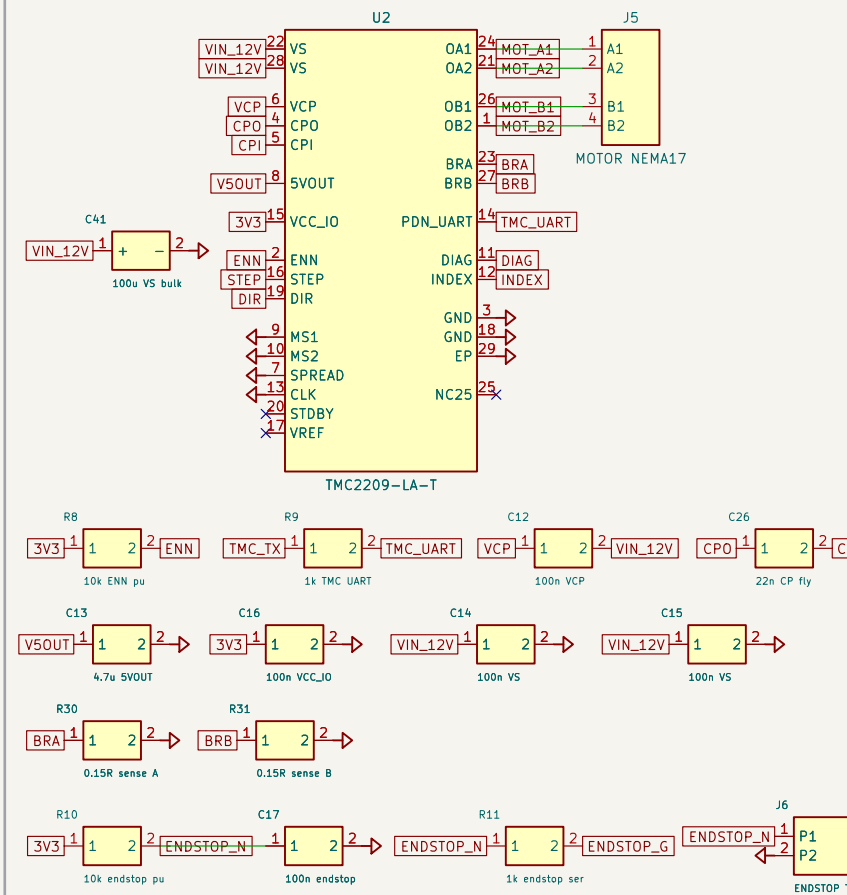
3. USB-C PROGRAMMING (data only; board powers from 12V)



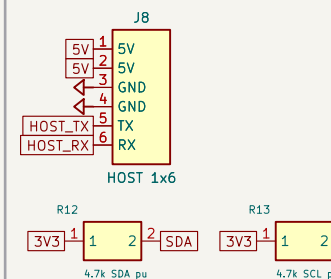
4. ESP32-S3-WROOM-1 (native USB; pin map ARCHITECTURE.md)



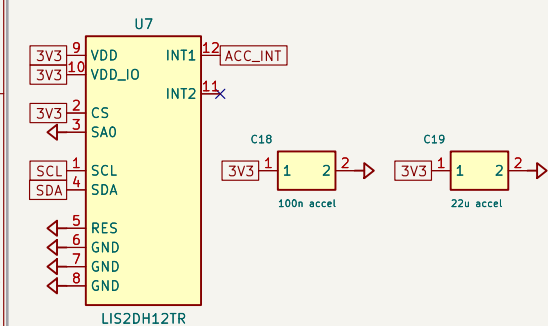
5. TMC2209: quiet StealthChop, UART, MOTOR OFF AT BOOT (R8) [ADR-0002]



6. HOST HEADER: 5V/1.5A MAX + UART; I2C bus pullups [D8]



7. LIS2DH12 ACCEL: lid angle ≥ 20 deg rule [D4]



8. CAPACITIVE: 4x MPR121QR2, 24 ELECTRODE LINES (short stubs!) [D4]

